RAJALAKSHMI ENGINEERING COLLEGE

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Y2

Started on	Wednesday, 30 September 2026, 9:33 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 9:34 AM
Time taken	55 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format
First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;

    // Read the target number n
    if (scanf("%d", &n) != 1) {
        return 0;
    }

    // Base cases handled directly
    if (n < 0) {
        printf("0\n");
        return 0;
    }
    if (n == 0 || n == 1 || n == 2) {
        printf("1\n");
        return 0;
    }

    // State variables representing: Ways(i-3), Ways(i-2), and Ways(i-1)
    long long ways_minus_3 = 1; // Ways(0)
    long long ways_minus_2 = 1; // Ways(1)
    long long ways_minus_1 = 1; // Ways(2)
    long long current_ways = 0;

    // Transition iteratively up to n
    for (int i = 3; i <= n; i++) {
        current_ways = ways_minus_1 + ways_minus_3;

        // Shift values forward for the next iteration
        ways_minus_3 = ways_minus_2;
        ways_minus_2 = ways_minus_1;
```

```
        ways_minus_1 = current_ways;
    }

    // Print the final result
    printf("%lld\n", current_ways);

    return 0;
}
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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